

THREAT REPORT: SHADOW-EARTH-066, EARTH DAHU

Summary

Assessed with high confidence per the source, the threat actor SHADOW-EARTH-066, also known as Earth Dahu, has been exploiting a vulnerability in WinRAR, specifically CVE-2025-8088, to target government and defense sectors in Ukraine. The malware families GIFTEDCROOK and GammaSteel have been identified in these attacks. Priority should be given to patching the affected systems and auditing Windows Startup folders and Run/RunOnce registry keys for unauthorized entries. The observed attacks highlight the importance of managing software updates to prevent such exploits.

Threat Overview

The threat actor, SHADOW-EARTH-066, Earth Dahu, has been utilizing the malware families GIFTEDCROOK and GammaSteel to exploit the vulnerability CVE-2025-8088. The primary targets of these attacks are government and defense sectors in Ukraine. The exploited vulnerability is a significant concern, as it allows the threat actor to gain unauthorized access to systems.

Attack Chain and Priority Control

The attack chain involves a series of techniques, including stealing web session cookies, using malicious files, and symmetric cryptography, ultimately leading to the exfiltration of data over a C2 channel. The priority technique in this chain is T1547.001 Registry Run Keys / Startup Folder, which is the graph chokepoint. To mitigate this threat, the priority action is to: Patch CVE-2025-8088 on all affected systems as the top priority, then: Audit Windows Startup folders and Run/RunOnce registry keys for unauthorized entries; alert on .lnk or script creation in Startup by browser/email temp paths.

Technical Appendix

Ground-truth telemetry from the source pulse; analytical scores are secondary and clearly labelled.

Observed Techniques (ATT&CK, expert-tagged by source)

- T1539 Steal Web Session Cookie
- T1204.002 Malicious File
- T1573.001 Symmetric Cryptography
- T1566.001 Spearphishing Attachment
- T1005 Data from Local System
- T1036 Masquerading
- T1555.003 Credentials from Web Browsers

- T1497 Virtualization/Sandbox Evasion
- T1041 Exfiltration Over C2 Channel
- T1059.001 PowerShell
- T1547.001 Registry Run Keys / Startup Folder
- T1027 Obfuscated Files or Information
- T1485 Data Destruction
- T1070.004 File Deletion
- T1071.001 Web Protocols
- T1564.004 NTFS File Attributes

Behavioural Attribution (secondary — NOT identity)

- Method: technique-overlap cosine vs 170 MITRE Groups, threshold tau=0.6.
- Result: **UNKNOWN_ADVERSARY** (top cosine 0.5488; below threshold).
- Nearest profiles: Inception (0.5488), RedCurl (0.5164), Nomadic Octopus (0.4824), Higaisa (0.4685), Stealth Falcon (0.4588)

Analytical Scores

- Priority technique (graph chokepoint): T1547.001 Registry Run Keys / Startup Folder (centrality 0.3042).
- Operational risk: Critical (score 1.0, reaches impact).

Indicators of Compromise (defanged — non-clickable)

The network and file indicators observed in this activity are listed below for blocklisting:

Type	Indicator
IP	166[.]0[.]132[.]237
IP	38[.]225[.]209[.]229
IP	23[.]26[.]237[.]80
IP	136[.]0[.]141[.]138
IP	194[.]58[.]66[.]82
IP	136[.]0[.]141[.]112
IP	136[.]0[.]141[.]41
IP	38[.]225[.]209[.]122
Domain	astrocaf[.]com
Domain	astrocaf[.]com

Type	Indicator
Domain	malicious[.]workers[.]dev
Domain	joymobile[.]com[.]ua
URL	hxxps://136[.]0[.]141[.]138:8406/rcv/
Hash	2a8ea9f1ad8936fb302243faa64b91c5767df411923715cbdb1a869e3bfd7e6d
Hash	c0b73ff43312d442260328a8cefd3b6
Hash	4528d5cf07bf0e1ac769b390236cab1bf34b938c
Hash	7200a9f1e1ea51b66ab9c9274e9d8f805633179634e8ff4dcb8ef82bc02518df
Hash	2af0a6135df3502a7f6de4d2de6db73b
Hash	b1c4a94df23638d70dae45f3193a64a6b036056d
Hash	3d371ef71e40c34a75c168d4647db096c2f386499d99a88d4e16b63cd4acda25
Hash	3d371ef71e40c34a75c168d4647db096c2f386499d99a88d4e16b63cd4acda25
Hash	a84375d4bd67c46d50fef7f7af31c7fb
Hash	526833a16669a85f0546809bfc35122e6f0bc17b
Hash	1c170b7470d507378ddb78e9d66305f1184e965baaf2d27ededb23a318a58953
Hash	2d9adb7932b7842dfb0e0f453b87e5d28dd4552094105e6340bad009956d8c2b
Hash	378809699c7252dc38b31969b9cc40858397759f15d6e418246dfaba9088fdd1
Hash	37b42a83715f7a34e00d3458d4f4b6e53b8c95372677ce020a2e38e80e60ba87
Hash	3c0330f9f934f86b6b70e108ff279a009b88a4a815acbed4adb3664e322e3e59
Hash	44f6f7ba668fc645129d66353e6f60402822ae929ce54648cae0bba6348a18ea
Hash	4e21c4c97aeb391473ee1e44961676f32de2ee8b56ecb136c1d8081df97c3db4
Hash	507b2fcdae058cebbd550965b90c44e878d7a2463058c846eeb68f0dc1b48eda
Hash	6083aac5376b7ca74cc363e0d66f70beaffee543d098c612b820b16fbfb0aa52
Hash	65c053030558b4a3588e2590c5c4961a9912180b731686deb3f4c831e765a095
Hash	718465f44c0680740fb61790eda3d2f4c5218c9de0c560299c580fa1602dc9c7
Hash	77963398e2c5c2fd9d28d9c5f9c2791cfbf422ba02225e01635dd7f5b31eff8

Type	Indicator
Hash	7d3ba419751e5ea52b567e1162f6a366bf3d06c44c8956a9f14520e9fb6ed0b1
Hash	8150b2b39fa62fa2de177ed8526c621a3581c0eb481dd9740fc5894ce2b7c13b
Hash	82fda6ea769d61aba230c3487787087cec53dd378e22f22a8fb8f0bd5ae83ded
Hash	89d20418450b34efe698bd36214100cfa49f60adf1c39a8bc8d65991b1ce2c23
Hash	a717dd74c01fcfce35a28f374e1c6f9ded06d6f7b0cc04618ce9454ad64febb8

